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CLAW DEVICE FOR CRANE MACHINE, GRIPPER FOR CRANE MACHINE, AND CRANE MACHINE

Abstract

A claw device for a crane machine, a gripper for a crane machine, and a crane machine. The gripper includes an adjustment assembly, an actuating assembly, and a limiting column. The adjustment assembly includes a fixed disk and connecting rods. The actuating assembly includes a pivot plate and clamping rods. Each clamping rod includes a connecting segment, a fixed segment, and a holding segment. Each connecting rod is connected to the fixed disk and the connecting segment. The limiting column defines a limiting hole. A lower part of the fixing hole corresponds to the limiting column, and an aperture of the fixing hole is greater than an aperture of the limiting hole; or another the limiting column is arranged corresponding to a lower side of the second adjustment hole, and an aperture of the second adjustment hole is greater than an aperture of the limiting hole.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation-in-part of U.S. application Ser. No. 18/812,296, which is filed on Aug. 22, 2024, the entire contents of which are hereby incorporated by reference in its entirety.

TECHNICAL FIELD
[0002] The present disclosure relates to the technical field of gaming equipment, and more specifically to a claw device for a crane machine, a gripper for a crane machine, and a crane machine.
BACKGROUND
[0003] In a capture game of catching or trapping a target, the game player needs to use a gripping claw to capture the game target. For different types of players, the claws should be designed to stimulate the interest of the players. In order to increase the challenge of the games, to stimulate the enthusiasm of adult gamers, the design of the claws should be aimed to achieve a relatively low success rate of the capture. Since child game players' operating ability is limited, in order to ensure that children can exercise hand and foot coordination and enjoy the fun in the games, the design of the claws should be aimed to achieve a relatively high success rate of the capture.
[0004] However, the existing claw design only meets the needs of adult gamers, and seldom considers the needs of child gamers. When the child game players use the claws designed for adult game players, the low success rate of the capture does not allow the child game players to obtain positive feedback in the games, which strikes the enthusiasm of the child game players to participate in the capture games. With the development of the crane machine market for children, the above problem is required to be mitigated or even solved.
SUMMARY OF THE DISCLOSURE
[0005] In view of the above problems, the present disclosure proposes a claw device for a crane machine, aiming to solve the technical problem that conventional claw devices are prone to tilting, and the low success rate of grabbing affects the sense of game experience of children gamers.
[0006] A claw device for a crane machine, including an adjustment assembly and an actuating assembly movably connected to the adjustment assembly; wherein the adjustment assembly includes a fixed disk and at least two connecting rods spaced apart along an outer peripheral edge of the fixed disk; the actuating assembly includes a pivot plate disposed below the fixed disk and at least two clamping rods spaced apart along an outer peripheral edge of the pivot plate; a viewing window is formed between each adjacent two of the at least two clamping rods; each clamping rod includes a connecting segment, a fixed segment, and a holding segment that are sequentially connected; the connecting segment is movably connected to the pivot plate; the fixed segment includes a first clamping surface toward a center axis of the pivot plate; the holding segment extends toward the center axis of the pivot plate from a lower end of the fixed segment; the holding segment includes a first holding surface toward the center axis of the pivot plate; the first holding surface is connected to the first clamping surface, and the first holding surface and the first clamping surface are arranged at an angle; an end of each connecting rod is movably connected to the fixed disk, and the other end of the connecting rod is movably connected to the connecting segment of a corresponding clamping rod; wherein the fixed disk defines a first adjustment hole

and a fixing hole, and the first adjustment hole and the fixing hole both pass through the fixed disk in an up-down direction; the pivot plate defines a second adjustment hole, and the second adjustment hole passes through the pivot plate in the up-down direction; a gravity center of the fixed disk is located in the first adjustment hole, and a projection of the gravity center of the fixed disk falls within the second adjustment hole; wherein the claw device further includes a limiting column; the limiting column defines a limiting hole, and the limiting hole runs through the limiting column in the up-down direction; the limiting hole is arranged with a mounting notch, and a width of the mounting notch is less than an inner diameter of the limiting hole; wherein, a lower part of the fixing hole corresponds to the limiting column, and an aperture of the fixing hole is greater than an aperture of the limiting hole; or another the limiting column is arranged corresponding to a lower side of the second adjustment hole, and an aperture of the second adjustment hole is greater than an aperture of the limiting hole.
[0007] In some embodiments, a lower surface of the fixed disk defines a first accommodation groove; a lower end of the fixing hole runs through the first accommodation groove, and the limiting column is arranged in the first accommodation groove; a sidewall surface of the first accommodation groove abuts against an outer peripheral surface of the limiting column and closes or seals the mounting notch of the limiting column.
[0008] In some embodiments, a lower surface of the pivot plate defines a second accommodation groove; a lower end of the second adjustment hole runs through the second accommodation groove, and the other limiting column is arranged in the second accommodation groove; a sidewall surface of the second accommodation groove abuts against an outer peripheral surface of the other limiting column and closes or seals the mounting notch of the other limiting column.
[0009] A gripper for a crane machine, including: an adjustment assembly, including a fixed disk and at least two connecting rods arranged at intervals along an outer periphery of the fixed disk; wherein the fixed disk defines a first adjustment hole and a fixing hole, and the first adjustment hole and the fixing hole both extend along an up-down direction and through the fixed disk; the first adjustment hole and the fixing hole are configured for a bead chain to move up and down; an actuating assembly, movably connected to the adjustment assembly; wherein the actuating assembly includes a pivot plate arranged below the fixed disk and at least two clamping rods arranged at intervals along an outer periphery of the pivot plate; a middle portion of the pivot plate defines a second adjustment hole extending along the up-down direction and through the pivot plate, and the second adjustment hole is configured for the bead chain to move up and down; an end of each connecting rod is movably connected to the fixed disk, and another end of the connecting rod is movably connected to a corresponding clamping rod; an end of each clamping rod is movably connected to the pivot plate; tail ends of each adjacent two of the at least two clamping rods are capable of moving together and of moving separated to achieve an opening-closing function of the gripper; and a limiting sleeve; wherein the limiting sleeve defines an opening on an end along an axial direction of the limiting sleeve and is arranged with an end wall at an opposite end; the end wall is arranged with a first channel, and a side wall of the limiting sleeve is arranged with a second channel communicating with the first channel; wherein the limiting sleeve is arranged below the fixing hole, a diameter of the fixing hole being greater than a maximum width of the first channel of the limiting sleeve; or, another the limiting sleeve is arranged below the second adjustment hole, a diameter of the second adjustment hole being greater than a maximum width of the first channel of the other limiting sleeve.
[0010] In some embodiments, an inner wall of the first channel is arranged with an anti-disconnecting member protruding from a middle position along an extension direction of the first channel, to prevent the bead chain from disengaging from the limiting sleeve.
[0011] In some embodiments, the anti-disconnecting member comprises two limiting protrusions arranged opposite each other, and the two limiting protrusions are arranged at intervals to define a slit; the two limiting protrusions and an inner wall surface of the first channel enclose to define a

mounting hole adapted to the bead chain; a width of the slit is less than a diameter of the mounting hole, and the slit is connected to the mounting hole and the second channel.
[0012] In some embodiments, an inner surface of the end wall defines a recess, and the first channel is disposed in the recess.
[0013] In some embodiments, an outer side wall of the limiting sleeve is arranged with a first snap-fit portion, and the fixed disk or the pivot plate is arranged with a second snap-fit portion that is adapted to the first snap-fit portion; one of the first snap-fit portion and the second snap-fit portion is a convex portion, and the other of the first snap-fit portion and the second snap-fit portion is a convex portion is a groove.
[0014] In some embodiments, the limiting sleeve is arranged with the second channel and an avoidance notch on opposite sides along the axial direction; the outer side wall of the limiting sleeve includes a guide bevel, and the avoidance notch is disposed on the guide bevel; the first snap-fit portion includes at least two protrusions, and the at least two protrusions are disposed on opposite sides of the second channel or the avoidance notch.
[0015] In some embodiments, a width of the second channel is less than a width of the first channel.
[0016] In some embodiments, the gripper further includes a pivot structure including a fixing post and a deformable part connected to each other, and the fixing post and the deformable part are arranged along an axial direction of the fixing post; wherein the fixing post and the deformable part pass through a corresponding connecting rod and the fixed disk, and the deformable part as well as the corresponding connecting rod or the fixed disk are limited in the axial direction of the fixing post; or, the fixing post and the deformable part pass through a corresponding connecting rod and a corresponding clamping rod, and the deformable part as well as the corresponding connecting rod or the corresponding clamping rod are limited in the axial direction of the fixing post; or, the fixing post and the deformable part pass through a corresponding clamping rod and the pivot plate, and the deformable part as well as the corresponding clamping rod or the pivot plate are limited in the axial direction of the fixing post.
[0017] In some embodiments, the pivot structure further includes two opposing bosses, and a pivot tab sandwiched between the two bosses; the two bosses and the pivot tab each define a through hole; the deformable part includes at least two elastic catches arranged on an end of the fixing post, and the at least two elastic catches are arranged along the axial direction of the fixing post; the fixing post is inserted into the through hole, and the at least two elastic catches are engaged with the two bosses in the axial direction of the fixing post; wherein one of a corresponding connecting rod and the fixed disk is arranged with the two bosses arranged opposite each other, and the other of the corresponding connecting rod and the fixed disk is arranged with the pivot tab; or, one of a corresponding connecting rod and the clamping rod is arranged with the two bosses, and the other of the corresponding connecting rod and the clamping rod is arranged with the pivot tab; or, one of a corresponding clamping rod and the pivot plate is arranged with the two bosses, and the other of the corresponding clamping rod and the pivot plate is arranged with the pivot tab.
[0018] In some embodiments, an end of the fixing post away from the at least two elastic catches extends beyond a surface of the two bosses in the axial direction of the fixing post.
[0019] In some embodiments, the fixing post has an internal cavity.
[0020] In some embodiments, each connecting rod and a corresponding clamping rod are rotatably connected; an end of the connecting rod close to the corresponding clamping rod is arranged with a first abutment protrusion, and a second abutment protrusion adapted to the first abutment protrusion is arranged on the corresponding clamping rod; the second abutment protrusion is arranged relative to the first abutment protrusion close to a central axis of the pivot plate; the second abutment protrusion is located in a movement path of the first abutment protrusion, and the first abutment protrusion is caused to be rotatable to abut against the second abutment protrusion.
[0021] In some embodiments, the gripper further includes a pallet that is removably mounted on

the tail end of each clamping rod; the pallet includes a third clamping surface disposed towards a central axis of the pivot plate and a third holding surface disposed opposite the third clamping surface; the third holding surface is attached on a plane of the clamping rod that is close to the central axis of the pivot plate; the pallet further includes an enclosure that is arranged around an edge of the third clamping surface.

[0022] In some embodiments, the gripper further includes a pallet that is removably mounted on the tail end of each clamping rod; the pallet includes a third clamping surface disposed towards a central axis of the pivot plate and a third holding surface disposed opposite the third clamping surface; the third holding surface is attached on a plane of the clamping rod that is close to the central axis of the pivot plate; the third holding surface is arranged with a limiting catch and an assembly groove; the limiting catch and the assembly groove are distributed along an extension direction of the clamping rod; a rod body of the clamping rod is engaged with the limiting catch, and the tail end of the clamping rod is inserted into the assembly groove.

[0023] In some embodiments, the gripper further includes a pallet that is removably mounted on the tail end of each clamping rod; an end of the pallet includes a second tooth that is protruding towards a central axis of the pivot plate.

[0024] In some embodiments, a through hole is defined on a middle portion of each clamping rod, and an aperture of the through hole extends along an extension direction of the clamping rod.

[0025] A crane machine, including a bead chain, a driving device, and the gripper as above; wherein the bead chain connects the gripper and the driving device; the driving device is configured to cause the fixed disk and the pivot plate of the gripper to move closer to or away from each other by driving the bead chain; in a case where the fixed disk and the pivot plate of the gripper move closer to each other, the at least two clamping rods of the gripper move closer to each other; in a case where the fixed disk and the pivot plate of the gripper move away from each other, the at least two clamping rods of the gripper move away from each other.

[0026] In summary, the present disclosure discloses a claw device for a crane machine. The claw device may limit the range of movement of the fixing rope or adjustment rope by making the aperture of the limiting hole in the limiting column less than the aperture of the fixing hole or the second adjustment hole, so as to avoid the success rate of grabbing falling due to the excessive tilting of the fixing rope or adjustment rope. In addition, the width of the mounting notch of the claw device is less than the diameter of the limiting hole, such that the fixing rope or adjustment rope can move within the limiting hole but cannot easily detach from the limiting column, thereby preventing the fixing rope and adjustment rope from detaching from the fixed disk or pivot plate when they are violently shaken to untangle the entanglement.

[0027] The present disclosure further discloses a gripper for a crane machine. The gripper may, through the setting of the limiting sleeve, solve the problem of the gripper tilting easily and the success rate of grabbing objects being not high due to the excessively large diameter of the fixing hole or the second adjustment hole.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] To illustrate the technical solutions of the embodiments of the present disclosure more clearly, the accompanying drawings of the embodiments will be briefly described below, and it will be apparent that the accompanying drawings in the following description relate only to some embodiments of the present disclosure and other drawings may be obtained from these drawings by those skilled in the art without creative labor.

[0029] FIG. **1** is a structural schematic view of a claw device for a crane machine according to some embodiments of the present disclosure.

[0030] FIG. **2** is a structural schematic view of a claw device for a crane machine according to other embodiments of the present disclosure.
[0031] FIG. **3** is an exploded structural schematic view of the claw device in FIG. **2**.
[0032] FIG. **4** is a structural schematic view of a claw device for a crane machine according to further other embodiments of the present disclosure.
[0033] FIG. **5** is a structural schematic view of a claw device for a crane machine according to still other embodiments of the present disclosure.
[0034] FIG. **6** is a structural schematic view of a claw device for a crane machine according to still other embodiments of the present disclosure.
[0035] FIG. **7**A illustrates a structural schematic view of a limiting column in FIG. **6** at one viewing angel. FIG. **7**B illustrates a structural schematic view of the limiting column in FIG. **6** at another viewing angle. FIG. **7**C illustrates a structural schematic view of the limiting column in FIG. **6** at another viewing angle.
[0036] FIG. **8**A illustrates a structural schematic view of a fixed disk in FIG. **6** at one viewing angel. FIG. **8**B illustrates a structural schematic view of the fixed disk in FIG. **6** at another viewing angle.
[0037] FIG. **9** is a structural schematic view of a fixed disk according to some embodiments of the present disclosure.
[0038] FIG. **10** is a structural schematic view of a gripper for a crane machine according to some embodiments of the present disclosure at a first viewing angle.
[0039] FIG. **11** is a structural schematic view of the gripper for the crane machine in FIG. **10** at a second viewing angle.
[0040] FIG. **12** is a structural schematic view of the gripper for the crane machine in FIG. **10** at a third viewing angle, with a pallet omitted.
[0041] FIG. **13** is a partial cross-sectional schematic view of a gripper for a crane machine according to some embodiments of the present disclosure at a viewing angle.
[0042] FIG. **14** is a partial cross-sectional schematic view of a gripper for a crane machine according to some embodiments of the present disclosure at another viewing angle.
[0043] FIG. **15** is a structural schematic view of a fixing member according to some embodiments of the present disclosure.
[0044] FIG. **16** is a structural schematic view of a limiting sleeve according to some embodiments of the present disclosure at a viewing angle.
[0045] FIG. **17** is a structural schematic view of a limiting sleeve according to some embodiments of the present disclosure at another viewing angle.
TABLE-US-00001 No. Name No. Name No. Name 100 Claw device for 13 First tab 224 First groove crane machine 10 Adjustment 14 First 23 Viewing window assembly rotating shaft 11 Fixed disk 20 Actuating 24 Second tab assembly 1111 First adjustment 21 Pivot plate 25 Second hole rotating shaft 112 Edge region 211 Second 30 Pallet adjustment hole 1121 Fixing hole 22 Clamping rod 31 Clamping portion 113 Connecting 221 Connecting 311 Second clamping channel segment surface 12 Connecting rod 222 Fixed segment 32 Holding portion 121 First 2221 First clamping 321 Second holding shaft hole surface surface 122 First notch 223 Holding segment 33 Mounting seat 123 Second groove 2231 First 331 Mounting groove holding surface 40 Rope 124 Second shaft hole 125 Second notch 50 Limiting column 51 Limiting hole 52 Mounting notch 61 First 62 Second 63 Fixing rope accommodation accommodation groove groove 64 Adjustment rope 65 First 66 Second limiting ball limiting ball 67 Third accommodation groove
[0046] The realization of the purpose, functional features, and advantages of the present disclosure will be further described in conjunction with the embodiments and with reference to the accompanying drawings.
DETAILED DESCRIPTION
[0047] The technical solutions in the embodiments of the present disclosure will be described

clearly and completely in the following in conjunction with the accompanying drawings in the embodiments of the present disclosure, and it is obvious that the described embodiments are only a part of the embodiments of the present disclosure and not all of the embodiments. Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without creative labor fall within the scope of the present disclosure. In addition, the technical solutions between various embodiments can be combined with each other, but it must be based on the fact that those skilled in the art is able to realize it. When the combination of the technical solutions appears to be contradictory or unattainable, it should be considered that the combination of such technical solutions does not exist, and is not included in the scope claimed by the present disclosure.

[0048] It should be noted that when the embodiments of the present disclosure involve directional indications (such as up, down, left, right, forward, back . . .), the directional indications are only intended to explain a relative positional relationship, a movement, etc. between the various components in a particular attitude. When the particular attitude changes, the directional indications are also changed accordingly.

[0049] In addition, when the embodiments of the present disclosure contain descriptions involving “first”, “second”, etc., the descriptions of “first”, “second”, etc. are intended only for descriptive purposes, and are not to be construed as indicating or implying their relative importance or implicitly specifying the number of the indicated technical features. That is, a feature defined as “first” or “second” may include at least one such feature either explicitly or implicitly. In addition, the meaning of “and/or” in the whole text is to include three concurrent solutions. For example, “A and/or B” includes an A solution, a B solution, and a solution in which A and B are satisfied at the same time.

[0050] The present disclosure proposes a claw device for a crane machine.

[0051] The claw device **100** includes an adjustment assembly **10** and an actuating assembly **20** movably connected to the adjustment assembly **10**; the adjustment assembly **10** includes a fixed disk **11** and at least two connecting rods **12** spaced apart along an outer peripheral edge of the fixed disk **11**; the actuating assembly **20** includes a pivot plate **21** disposed below the fixed disk **11** and at least two clamping rods **22** spaced apart along an outer peripheral edge of the pivot plate **21**; a viewing window **23** is formed between adjacent two clamping rods **22**; each clamping rod **22** includes a connecting segment **221**, a fixed segment **222**, and a holding segment **223** that are sequentially connected; the connecting segment **221** is movably connected to the pivot plate **21**; the fixed segment **222** includes a first clamping surface **2221** toward a center axis of the pivot plate **21**; the holding segment **223** extends toward the center axis of the pivot plate **21** from a lower end of the fixed segment **222**; the holding segment **223** includes a first holding surface **2231** toward the center axis of the pivot plate **21**; the first holding surface **2231** is connected to the first clamping surface **2221**, and the first holding surface **2231** and the first clamping surface **2221** are arranged at an angle; an end of the connecting rod **12** is movably connected to the fixed disk **11**, and the other end of the connecting rod **12** is movably connected to the connecting segment **221**.

[0052] The claw device **100** refers to a component of the crane machine that can simulate an opening or closing action of a hand to grasp a subject (e.g., doll). The adjustment assembly **10** refers to a component of the claw device **100** that can drive other components to move to accomplish the opening or closing action. The actuating assembly **20** refers to a component of the claw device **100** that directly accomplishes the opening or closing action. The adjustment assembly **10** includes a fixed disk **11** and at least two connecting rods **12** spaced along an outer peripheral edge of the fixed disk **11**; the number of the connecting rods **12** may be two, three, four, etc. The actuating assembly **20** includes a pivot plate **21** disposed below the fixed disk **11** and at least two clamping rods **22** spaced along an outer peripheral edge of the pivot plate **21**. The number of the clamping rods **22** is set corresponding to the number of the connecting rods **12**; the number of the clamping rods **22** may be two, three, four, etc.

[0053] A viewing window **23** is formed between each adjacent two clamping rods **22**. Compared to a clamping member shaped like a bucket, the rod-shaped clamping member in the embodiments may facilitate the game player to observe the positional relationship between a target subject and a clamping portion **31**, and thus assist the game player in adjusting the game strategy to improve the grasping success rate.

[0054] The clamping rod **22** includes a connecting segment **221**, a fixed segment **222**, and a holding segment **223** that are sequentially connected. The connecting segment **221** is movably connected to the pivot plate **21**. An end of the connecting segment **221** is movably connected to the pivot plate **21**, and the adjustment assembly **10** is configured to drive the connecting segment **221** to move by adjusting a height of the pivot plate **21**; the other end of the connecting segment **221** is connected to the fixed segment **222**. The fixed segment **222** includes a first clamping surface **2221** toward a center axis of the pivot plate **21**; the holding segment **223** extends from a lower end of the fixed segment **222** toward the center axis of the pivot plate **21**; the holding segment **223** includes a first holding surface **2231** toward the center axis of the pivot plate **21**; the first holding surface **2231** is connected to the first clamping surface **2221**, and the first holding surface **2231** and the first clamping surface **2221** are arranged at an angle. In some embodiments, the first holding surface **2231** is flat. In this way, when the target subject is being grasped, the first holding surface **2231** can uniformly provide a supporting force, such that the target subject will not fall down due to an unbalanced force during the movement of the claw device.

[0055] The angle between the first holding surface **2231** and the first clamping surface **2221** is greater than or equal to 90 degrees and less than or equal to 120 degrees. If the angle between the first holding surface **2231** and the first clamping surface **2221** is less than 90 degrees, the first holding surface **2231** can provide a limited support force to the target subject, making it difficult to secure the target subject in the claw device. If the angle between the first holding surface **2231** and the first clamping surface **2221** is greater than 120 degrees, the target subject is very likely to fall when the claw device **100** is moving. In the embodiments, however, the angle between the first holding surface **2231** and the first clamping surface **2221** is greater than or equal to 90 degrees and less than or equal to 120 degrees. When grasping the target subject, this angle may make the target subject be fixed in the claw device to prevent the claw device being failed in grasping; when the claw device is moving, this angle may ensure that the first holding surface **2231** can provide sufficient upward support force to prevent the target subject from falling.

[0056] An end of the connecting rod **12** is movably connected to the fixed disk **11**, and the other end of the connecting rod **12** is movably connected to the connecting segment **221**. The claw device **100** can cause the connecting rod **12** to move relative to the fixed disk **11** and the pivot plate **21** by adjusting a height difference between the fixed disk **11** and the pivot plate **21**. Since the connecting rod **12** is movably connected to the connecting segment **221**, the connecting rod **12** can drive the connecting segment **221** to move, which in turn causes the multiple clamping members to separate from each other or come close to each other, thereby accomplishing the simulated opening or closing action of the hand.

[0057] The present disclosure discloses a claw device **100** for a crane machine, including an adjustment assembly **10** and an actuating assembly **20**; the adjustment assembly **10** includes a fixed disk **11** and connecting rods **12**; the actuating assembly includes a pivot plate **21** and clamping rods **22**; an viewing window **23** is formed between each adjacent two of the clamping rods **22**; each clamping rod **22** includes a connecting segment **221**, a fixed segment **222**, and a holding segment **223** that are sequentially connected, where the connecting segment **221** is movably connected to the pivot plate **21**, and the holding segment **223** extends towards a center axis of the pivot plate; a first holding surface **2231** of the holding segment **223** is connected to a first clamping surface **2221** of the fixed segment **222**, and the first holding surface **2231** and the first clamping surface **2221** are arranged at an angle; an end of the connecting rod **12** is connected to the fixed disk **11**, and the other end of the connecting rod **12** is connected to the connecting segment **221**. In the present disclosure, the first clamping surface **2221** and the first holding surface **2231** are arranged at an angle to enable the target subject to be firmly grasped by the claw device without easily falling, and the viewing window **23** formed between adjacent clamping rods **22** facilitates the game player to observe the positional relationship between the target subject and the clamping rods **22** to adjust the game strategy, thereby improving the grasping success rate of the claw device.

[0058] Further, referring to FIG. **4**, a width direction of the first holding surface **2231** intersects with an extension direction of the holding segment **223**; a width of the first holding surface **2231** gradually increases along the extension direction of the holding segment **223** away from the fixed segment **222**.

[0059] In the embodiments, the gradual increase in the width of the first holding surface **2231** along the extension direction of the holding segment **223** may increase the contact area between the first holding surface **2231** and the target subject, such that the target subject can be steadily grasped by the claw device. In other words, the increase in the contact area between the first holding surface **2231** and the target subject may facilitate the claw device to fix the target subject and further prevent the target subject from falling during the movement of the claw device.

[0060] Further, referring to FIG. **5**, the claw device **100** includes a pallet **30** detachably arranged on the fixed segment **222**. The pallet **30** includes a clamping portion **31** and a holding portion **32** connected to a lower side of the clamping portion **31**; the clamping portion **31** includes a second clamping surface **311** extending along a width direction of the first clamping surface **2221**; a width of the second clamping surface **311** is greater than a width of the first clamping surface **2221**. Referring to FIGS. **4** and **5**, the pallet **30** further includes the holding portion **32** connected to the lower side of the clamping portion **31**; the holding portion **32** includes a second holding surface **321** extending along a width direction of the first holding surface **2231**; a width of the second holding surface **321** is greater than a width of the first holding surface **2231**; a flat surface on the pallet that is back away from the second holding surface **321** is attached to the first holding surface **2231**.

[0061] In the embodiments, the claw device **100** includes the pallet **30** detachably arranged on the fixed segment **222**. The pallet **30** may be in a variety of shapes, such as triangle, circle, rectangle, and other regular and irregular shapes. The pallet **30** may be disposed both on an inner side and/or an outer side of the fixed segment **222**. The pallet **30** may increase the contact area with the target subject, enabling the claw device to securely clamp the target subject. The pallet **30** is detachably arranged on the holding segment **223**, which may facilitate parents to adjust the difficulty of the game by mounting or removing the pallet **30**, such that the crane machine game can be adapted to child gamers of different ages.

[0062] The pallet **30** includes a clamping portion **31**. The clamping portion **31** includes a second clamping surface **311** extending in a width direction of the first clamping surface **2221**; the width of the second clamping surface **311** is greater than the width of the first clamping surface **2221**. Compared to the first holding surface **2221**, the second holding surface **311** increases the contact area with the target subject, thereby providing sufficient support and friction at the position of the fixed segment **222**, such that the clamping rod **22** can clamp the target subject more securely.

[0063] Compared to the first holding surface **2231**, the second holding surface **321** increases the contact area with the target subject, thereby providing more upward support force at the position of the holding segment **223**, such that that the target subject may not be dislodged from the claw device easily.

[0064] A flat surface on the pallet that is back away from the second holding surface **321** is attached to the first holding surface **2231**. In other words, the pallet **30** is arranged on the inner side of the fixed segment **222**, rather than on the outer side of the fixed segment **222**. When the pallet **30** is arranged on the outer side of the fixed segment **222**, the fixed segment **222** is sandwiched between the target subject and the pallet **30**. When the pallet **30** is arranged on the inner side of the fixed segment **222**, the target subject is in direct contact with the pallet **30**, i.e., there is no barrier

between the target subject and the pallet **30**. When the target subject is filled with cotton, the grasping success rate of the claw device is not greatly affected by whether the pallet **30** is arranged on the inner side or the outer side of the fixed segment **222**. Currently, in order to increase the fun of the crane machine game, the target subject may further be a doll made of plastic or a doll arranged in a spherical mold, and the game player can directly grasp the spherical mold. When the pallet **30** is arranged on the outer side of the fixed segment **222**, the fixed segment **222** clamped between the target subject and the pallet **30** may easily cause the doll to fall off, thereby reducing the success rate of grasping by the claw device. In the embodiments, the pallet **30** is arranged on the inner side of the fixed segment **222**, and the pallet **30** can be in full contact with the target subject without obstruction, which may improve the grasping success rate of the claw device.
[0065] Further, referring to FIG. **5**, a partial region of the pallet **30** is raised to form a mounting seat **33**, and the mounting seat **33** defines a mounting groove **331**; an inner diameter of the mounting groove **331** is gradually decreased along a length direction of the clamping rod **22**, and the minimum inner diameter of the mounting groove **331** is less than or equal to an outer diameter of the fixed segment **222**; the fixed segment **222** is detachably arranged in the mounting groove **331**.
[0066] In the embodiments, a partial region of the pallet **30** is raised to form a mounting seat **33**, and the mounting seat **33** defines a mounting groove **331**. In some embodiments, a central region of the pallet **30** is raised to form the mounting seat **33**. Compared with other regions of the pallet **30** being raised to form the mounting seat **33**, the central region of the pallet **30** being raised to form the mounting seat **33** may enable the pallet **30** to uniformly withstand the tension from the target subject, preventing the pallet **30** from rotating around the fixed segment **222** due to uneven force on the pallet **30**. In addition, the central region of the pallet **30** being raised to form the mounting seat **33** makes the grasping stability of the claw device better, thereby improving the grasping success rate of the claw device, which is more suitable for a child gamer.
[0067] The inner diameter of the mounting groove **331** is gradually reduced along the length direction of the clamping rod **22**, and the minimum inner diameter of the mounting groove **331** is less than or equal to the outer diameter of the fixed segment **222**, which may facilitate the mounting of the fixed segment **222** to the mounting groove **331** from a position where the inner diameter of the mounting groove **331** is greater, and further enable the mounting groove **331** to have an interference fit with the fixed segment **222** at a position where the inner diameter of the mounting groove **331** is the least. Whether the fixed segment **222** can be stably assembled with the mounting groove **331** has a significant impact on the grasping success rate of the claw device. In the embodiments, the tapering of the inner diameter of the mounting groove **331** allows the mounting groove **331** and the fixed segment **222** to be securely assembled, which further improves the grasping success rate of the claw device.
[0068] In some embodiments, referring to FIGS. **1**, **2**, **4**, **6**, **8**, and **9**, the fixed disk **11** defines a first adjustment hole **1111** and a fixing hole **1121**, and the first adjustment hole **1111** and the fixing hole **1121** both pass through the fixed disk **11** in an up-down direction; the pivot plate **21** defines a second adjustment hole **211**, and the second adjustment hole **211** passes through the pivot plate **21** in the up-down direction; a gravity center of the fixed disk **11** is located in the first adjustment hole **1111**, and a projection of the gravity center of the fixed disk **11** falls within the second adjustment hole **211**.
[0069] During mounting, a fixing rope **63** passes through the fixing hole **1121** to realize fixing, and an adjustment rope **64** passes through the first adjustment hole **1111** and the second adjustment hole **211** in turn to realize adjustment. It is to be understood that when the claw device **100** is in operation, the adjustment rope **64** and the fixing rope **63** cooperate with each other to realize the movement and grasping-releasing action of the claw device **100**. In some embodiments, referring to FIG. **9**, a center axis of the second adjustment hole **211** and a center axis of the first adjustment hole **1111** are on the same straight line. Specifically, the fixed disk **11** includes a center region facing the center of the pivot plate **21** and an edge region **112** disposed along a periphery of the

center region; the edge region **112** defines the fixing hole **1121**; the center region defines the first adjustment hole **1111**; and the pivot plate **21** defines the second adjustment hole **211** facing the first adjustment hole **1111**. In some embodiments, the first adjustment hole **1111** and the second adjustment hole **211** are not in the same straight line. Specifically, the fixing hole **1121** and the first adjustment hole **1111** are located in the middle of the fixed disk **11**, and the second adjustment hole **211** faces the gravity center of the fixed disk **11**. The gravity center of the fixed disk **11** is located in the first adjustment hole **1111**, and the positive projection of the gravity center of the fixed disk **11** falls into the region of the second adjustment hole **211**, so as to cause the claw device **100** in the initial position to be in balance. In this way, it is easy for the game player to adjust the drop angle of the claw device **100** during the game, improving the grasping success rate.

[0070] Further, referring to FIGS. **7**A, **7**B, **7**C, and **8**A and **8**B, the claw device **100** further includes a limiting column **50**. The limiting column **50** defines a limiting hole **51**, and the limiting hole **51** runs through the limiting column **50** in the up-down direction. The limiting hole **51** is arranged with a mounting notch **52**, and the width of the mounting notch **52** is less than the inner diameter of the limiting hole **51**. A lower part of the fixing hole **1121** corresponds to the limiting column **50**, and the aperture of the fixing hole **1121** is greater than the aperture of the limiting hole **51**; or, another limiting column **50** is arranged corresponding to a lower side of the second adjustment hole **211**, and the aperture of the second adjustment hole **211** is greater than the aperture of the limiting hole **51**.

[0071] Currently, conventional crane machines often have the phenomenon of the fixing rope **63** and the adjustment rope **64** becoming entangled with each other when in use. For this reason, the gamer usually operates a controller to cause the claw device **100** to shake vigorously, thereby causing the fixing rope **63** and the adjustment rope **64** to untangle from each other. However, the violent shaking may cause the fixing rope **63** or the adjustment rope **64** to detach from the fixed disk **11** or the pivot plate **21**. In addition, in order to ensure that the adjustment rope **64** or the fixing rope **63** moves smoothly in the up-down direction, the aperture diameters of the fixing hole **1121**, the first adjustment hole **1111**, or the second adjustment hole **211** are provided large enough. Such apertures mean that the range of movement of the fixing rope **63** or the adjustment rope **64** is also wider. When the claw device **100** has a tendency to tilt or when tilting occurs, the offset of the fixing rope **63** or the adjustment rope **64** may contribute to the tilt or exacerbate the tilting effect.

[0072] In the embodiments, the lower end of the fixing rope **63** or the adjustment rope **64** can be arranged on the limiting column **50** from the mounting notch **52**. The width of the mounting notch **52** is less than the aperture of the limiting hole **51**, such that the fixing rope **63** or the adjustment rope **64** can be moved inside the limiting hole **51**, but cannot be easily detached from the limiting column **50**. In case of wear and tear of a part or damage of the part, the mounting notch **52** facilitates the adjustment rope **64** or the fixing rope **64** to be separated from the limiting column **50** for replacement.

[0073] In some embodiments, the limiting column **50** is arranged below the fixing hole **1121**, and the limiting column **50** may either not be connected to the fixed disk **11** or may be connected to the fixed disk **11**, such as bonded, threaded, snap-fit, and the like. The aperture of the limiting hole **51** is less than the aperture of the fixing hole **1121**, for limiting the range of movement of the fixing rope **63** and reducing the effect of the offset of the fixing rope **63** on the overall tilt of the claw device **100**. In some embodiments, the limiting column **50** is arranged below the second adjustment hole **211**, and the limiting column **50** may either not be connected to the pivot plate **21** or may be connected to the pivot plate **21**, such as bonded, threaded, snap-fit, and the like. The aperture of the second adjustment hole **211** is greater than the aperture of the limiting hole **51**, for limiting the range of movement of the adjustment rope **64** and reducing the effect of the offset of the adjustment rope **64** on the overall tilt of the claw device **100**. In some embodiments, the limiting column **50** is arranged below each of the fixing hole **1121** and the second adjustment hole **211**.

[0074] Further, referring to FIG. **6**, a lower surface of the fixed disk **11** defines a first accommodation groove **61**. The lower end of the fixing hole **1121** runs through the first accommodation groove **61**, and the limiting column **50** is arranged in the first accommodation groove **61**; a sidewall surface of the first accommodation groove **61** abuts against an outer peripheral surface of the limiting column **50** and closes or seals the mounting notch **52**; a lower surface of the pivot plate **21** defines a second accommodation groove **62**. The lower end of the second adjustment hole **211** runs through the second accommodation groove **62**, and another limiting column **50** is arranged in the second accommodation groove **62**; a sidewall surface of the second accommodation groove **62** abuts against the outer peripheral surface of the limiting column **50** and closes or seals the mounting notch **52**.

[0075] In the embodiments, the lower surface of the fixed disk **11** defines a first accommodation groove **61**, a lower end of the fixing hole **1121** runs through the first accommodation groove **61**, and the limiting column **50** is arranged in the first accommodation groove **61**, i.e., the lower end of the fixing rope **63** can pass through the fixing hole **1121**, the first accommodation groove **61**, and the limiting column **50** in turn, and be connected to the limiting column **50**. A sidewall surface of the first accommodation groove **61** abuts against an outer peripheral surface of the limiting column **50**, such that the limiting column **50** can be hidden and fixed in the first accommodation groove **61**. In this way, the overall structure of the claw device **100** is more compact. The mounting notch **52** closed or sealed on the sidewall surface of the first accommodation groove **61** prevents the fixing rope **63** from detaching from the limiting column **50** even when the claw device **100** is shaken violently.

[0076] The lower surface of the pivot plate **21** defines a second accommodation groove **62**, a lower end of the second adjustment hole **211** runs through the second accommodation groove **62**, and the second accommodation groove **62** is arranged with another limiting column **50**, i.e., the lower end of the adjustment rope **64** can pass through the first adjustment hole **1111**, the second adjustment hole **211**, the second accommodation groove **62**, and the limiting column **50** in turn and be connected to the limiting column **50**. The sidewall surface of the second accommodation groove **62** abuts against an outer peripheral surface of the limiting column **50**, such that the limiting column **50** can be hidden and fixed in the second accommodation groove **62**. In this way, the overall structure of the claw device **100** is more compact. The mounting notch **52** closed or sealed on the sidewall surface of the second accommodation groove **62** prevents the fixing rope **63** from detaching from the limiting column **50** even when the claw device **100** is shaken violently.

[0077] In some embodiments, the fixing rope **63** includes a first rope body and multiple first limiting balls **65** fixed to the first rope body and sequentially distributed along a length direction of the first rope body. The first rope body enters into the limiting hole **51** from the mounting notch **52**, and the aperture diameter of the limiting hole **51** is greater than the diameter of each first limiting ball **65**, such that an upward movement of the first rope body may be limited.

[0078] The adjustment rope **64** includes a second rope body and multiple second limiting balls **66** fixed on the second rope body and sequentially distributed along a length direction of the second rope body. The second rope body enters into the limiting hole **51** from the mounting notch **52**, and the aperture of the limiting hole **51** is greater than the diameter of each second limiting ball **66**, such that an end of the second rope body may be limited from disengaging from the pivot plate.

[0079] In some embodiments, the fixed disk **11** is further arranged with a connecting channel **113** that connects the fixing hole **1121** and the first adjustment hole **1111**, and the connecting channel **113** runs through the fixed disk **11** in the up-down direction.

[0080] During movement, the claw device **100** may sway due to inertia, causing the claw device **100** to be skewed. Or, although the claw device **100** grasps the target subject, the gravity center of the target subject is not on the center axis of the claw device **100**, causing the claw device **100** to be skewed. The skewing of the claw device **100** may easily cause the target subject to fall, reducing the grasping success rate of the claw device **100**. In the embodiments, when the claw device **100** is skewed, a rope **40** may be biased to an un-skewed side through the connecting channel **113** to keep the claw device **100** balanced, avoiding the target subject from falling due to the skewing of the claw device **100**. Taking the skewing of the claw device **100** towards the side with the fixing hole **1121** as an example, the adjustment rope **64** may be slightly skewed towards the first adjustment hole **1111** through the connecting channel **113** to restore the balance.

[0081] In some embodiments, the pivot plate **21** is arranged with a counterweight member, and the gravity center of the counterweight member coincides with the gravity center of the pivot plate **21** in the up-down direction.

[0082] The clamping rod **22** is pivotally connected to both the pivot plate **21** and the connecting rod **12**. When the clamping rods **22** are separated from each other, the clamping rods **22** are affected by friction, and the opening angle between neighboring clamping rods **22** cannot be achieved as desired. When encountering a target subject with a large size, the clamping rod **22** may not be able to grab the target subject due to the insufficient opening angle, thereby discouraging the game player. In the embodiments, the pivot plate **21** is arranged with a counterweight member, and the gravity center of the counterweight member coincides with the gravity center of the pivot plate **21** in the up-down direction, which uniformly increases the weight of the pivot plate **21**. In this way, the pivot plate **21** is further separated from the fixed disk **11** by gravity, thereby allowing the clamping rods **22** to open at a greater angle. In some embodiments, an upper surface of the pivot plate **21** defines a third accommodative groove **67**. The third accommodation groove **67** is arranged around the gravity center of the pivot plate **21**, and the counterweight member is arranged within the third accommodation groove **67**. In some embodiments, the second adjustment hole **211** is arranged at the gravity center of the pivot plate **21**, the third accommodation groove **67** is arranged around the second adjustment hole **211**, and the counterweight is arranged in the third accommodation groove **67**. In this way, it is possible to make the opening angle of the clamping rods **22** greater, and at the same time make the structure of the claw device **100** more compact.

[0083] In some embodiments, referring to FIGS. **2** and **3**, the adjustment assembly **10** includes at least two pairs of first tabs **13** spaced apart along the outer peripheral edge of the fixed disk **11** and a first rotating shaft **14** connecting each pair of first tabs **13**; the connecting rod **12** defines a first shaft hole **121** at an end near the fixed disk **11**, and a first notch **122** is arranged on an inner side of the connecting rod **12** corresponding to the first shaft hole **121**; the first notch **122** is in communication with the first shaft hole **121**; the first rotating shaft **14** is rotatably arranged in the first shaft hole **121** from the first notch **122**; or [0084] the actuating assembly **20** further includes two second tabs **24** disposed opposite each other on the connecting segment **221** and a second rotating shaft **25** connecting the two second tabs **24**; the connecting rod **12** defines a second shaft hole **124** at an end near the connecting segment **221**, and a second notch **125** is defined on an inner side of the connecting rod **12** corresponding to the second shaft hole **124**; the second notch **125** is in communication with the second shaft hole **124**; the second rotating shaft **25** is rotatably arranged in the second shaft hole **124** from the second notch **125**.

[0085] Conventionally, in the pivot structure, the rotating shaft is inserted into the shaft hole, and a locking member is arranged at an end of the rotating shaft. Due to the small size of the locking member, the mounting thereof usually requires external tools, which reduces assembly efficiency to some extent. In addition, during use, the locking member is prone to loosening. After the locking member is loosened, the connecting rod **12** is unable to drive the clamping rod **22** normally, resulting in a lower grasping success rate of the claw device. In the proposed embodiments, the first rotating shaft **14** and the second rotating shaft **25** realize the rotational connection with the fixed disk **11** and the connecting segment **221** from the first notch **122** and the second notch **125**, respectively. The assembly process is efficient and convenient without the use of external tools. In addition, the first tab **13** and the second tab **24** prevent the first shaft hole **121** and the second shaft hole **124**, respectively, from detaching from the rotating shaft. Further, the first tab **13** and the second tab **24** will not be loosened during use, such that the clamping rod **22** is enabled to work properly.

[0086] In some embodiments, referring to FIGS. **2** and **3**, each of sides of the clamping rod **22** adjacent to the first holding surface **2231** defines multiple first grooves **224** spaced apart along its length direction, or each of opposite sides of the connecting rod **12** defines multiple second grooves **123** spaced apart along its length direction.

[0087] Whether the claw device is balanced or not has a very important influence on the grasping success rate. Whether the mass of the clamping rod **22** and the connecting rod **12** is evenly distributed is crucial for the balance of the claw device. In some embodiments, the clamping rod **22** or the connecting rod **12** are made of plastic. Due to their certain thickness, the phenomenon of size shrinkage or hole shrinkage is likely to occur during manufacturing, resulting in uneven mass distribution of the clamping rod **22** or the connecting rod **12**, which in turn affects the balance of the claw device. In the proposed embodiments, the hollowing out of the glue may be realized by defining grooves on the clamping rod **22** or the connecting rod **12** to avoid uneven mass distribution caused by the poor surface of the clamping rod **22** and/or the connecting rod **12**, such that the balance of the claw device may not be affected.

[0088] The number of the second grooves **123** may be one, two or more. In some embodiments, the number of the second grooves **123** is three. The connecting rod **12** with three second grooves **123** may facilitate assembly by the visually impaired person. The visually impaired person can sense the number of the second grooves **123** by touching, so as to determine the position of the two ends of the connecting rod **12**. In some embodiments, a side of the connecting rod **12** has a first notch **122**. The visually impaired person can sense the number of the second notches **123** by touch and determine the position of the connecting rod **12** that he or she is holding relative to the position of the first notch. In this way, it is convenient for the visually impaired person to quickly mount the second rotating shaft **25** from the first notch **122** to the first shaft hole **121**.

[0089] The present disclosure further proposes a crane machine, the crane machine includes a claw device **100**, a driving device, and a rope **40**, and the specific structure of the claw device **100** is referred to the above embodiments. Since the crane machine is adopted with all the technical solutions of all the above embodiments, it has at least all the beneficial effects brought about by the technical solutions of the above embodiments, which will not be repeated herein. The rope **40** connects the claw device **100** and the driving device; the driving device is configured to cause the fixed disk **11** and the pivot plate **21** of the claw device to be close to each other or far away from each other by driving the fixing rope **63**; when the fixed disk **11** and the pivot plate **21** of the claw device are close to each other, the clamping rods **22** of the claw device are close to each other; when the fixed disk **11** and the pivot plate **21** of the claw device are far away from each other, the clamping rods **22** are far away from each other.

[0090] In some embodiments, referring to FIGS. **10** to **17**, a gripper for a crane machine includes a limiting sleeve **10***b*, an adjustment assembly **20***b*, and an actuating assembly **30***b*.

[0091] The limiting sleeve **10***b* defines an opening **11***b* on an end in the axial direction thereof and is arranged with an end wall **12***b* at the other end; the end wall **12***b* is arranged with a first channel **125***b*, and a side wall of the limiting sleeve **10***b* is arranged with a second channel **13***b* communicating with the first channel **125***b*.

[0092] The adjustment assembly **20***b* includes a fixed disk **21***b* and at least two connecting rods **22***b* arranged at intervals along an outer periphery of the fixed disk **21***b*; the fixed disk **21***b* defines a first adjustment hole **211***b* and a fixing hole **212***b*, both of which extend along an up-down direction and through the fixed disk **21***b*; the first adjustment hole **211***b* and the fixing hole **212***b* allow a bead chain to move up and down.

[0093] The actuating assembly **30***b* is movably connected to the adjustment assembly **20***b*; the actuating assembly **30***b* includes a pivot plate **31***b* arranged below the fixed disk **21***b* and at least two clamping rods **32***b* arranged at intervals along an outer periphery of the pivot plate **31***b*; the pivot plate **31***b* defines a second adjustment hole **311***b* arranged in the center and extending along the up-down direction and through the pivot plate **31***b*, and the second adjustment hole **311***b* allows the bead chain to move up and down; an end of the connecting rod **22***b* is movably connected to the fixed disk **21***b*, and the other end of the connecting rod **22***b* is movably connected to the clamping rod **32***b*; an end of the clamping rod **32***b* is movably connected to the pivot plate **31***b*, and tail ends of the adjacent clamping rods **32***b* are brought together or separated to open or close the gripper.

[0094] The limiting sleeve **10***b* is arranged below the fixing hole **212***b*, a diameter of the fixing hole **212***b* being greater than a maximum width of the first channel **125***b*, and the limiting sleeve **10***b* is engaged with the fixed disk **21***b*; or, another limiting sleeve **10***b* is arranged below the second adjustment hole **311***b*, the diameter of the second adjustment hole **311***b* being greater than the maximum width of the first channel **125***b*, and the limiting sleeve **10***b* is engaged with the pivot plate **31***b*.

[0095] In the above embodiments, the gripper of the crane machine refers to a part of the crane machine that can simulate the opening and closing of a hand to grab toys such as dolls and puppets.

[0096] The adjustment assembly **20***b* refers to a part, of the gripper of the crane machine, that can drive the movement of other parts to complete the opening or closing action. The adjustment assembly **20***b* includes a fixed disk **21***b* and at least two connecting rods **22***b* spaced along the outer periphery of the fixed disk **21***b*; the connecting rods **22***b* may be two, three, four, etc. in number. An end of the connecting rod **22***b* is movably connected to the fixed disk **21***b*, and the other end of the connecting rod **22***b* is movably connected to the connecting segment. The gripper of the crane machine moves relative to the fixed disk **21***b* and the pivot plate **31***b* by adjusting the height difference between the fixed disk **21***b* and the pivot plate **31***b*. Since the connecting rod **22***b* is movably connected to the connecting segment, the connecting rod **22***b* drives the connecting segment to move, thereby causing multiple clamping members to separate from each other or move closer to each other to complete the action of simulating the opening and closing of a hand.

[0097] The actuating assembly **30***b* refers to a part of the gripper **100***b* of the crane machine that directly performs the opening or closing action. The actuating assembly **30***b* includes a pivot plate **31***b* arranged below the fixed disk **21***b* and at least two clamping rods **32***b* spaced along the outer periphery of the pivot plate **31***b*. The number of the clamping rods **32***b* corresponds to the number of the connecting rods **22***b*, and the number of clamping rods **32***b* may be two, three, four, etc.

[0098] The limiting sleeve **10***b* is configured to limit the position of the bead chain to improve the success rate of gripping by the gripper. In some embodiments, the limiting sleeve **10***b* is only arranged below the fixing hole **212***b* or the second adjustment hole **311***b*; in other embodiments, several limiting sleeves **10***b* are arranged below the fixing hole **212***b* and the second adjustment hole **311***b*. The opening **11***b*, the second channel **13***b*, and the first channel **125***b* in the limiting sleeve **10***b* provide an installation path for the bead chain. The bead chain includes several beads and a string connecting the beads in series. The end wall **12***b* is configured to abut against the bead at one end of the bead chain, thereby restricting the upward movement of the bead out of the limiting sleeve **10***b*.

[0099] During the mounting process, the beads are placed in the opening **11***b* at an end of the limiting sleeve **10***b*, and the string between adjacent beads is aligned with the second channel **13***b*; the chain is moved along the second channel **13***b* until the beads abut against the end wall **12***b*; the beads are caused to move along the first channel **125***b*; the limiting sleeve **10***b* is arranged below the fixing hole **212***b* or the second adjustment hole **311***b*, such that the hole wall of the fixing hole **212***b* or the second adjustment hole **311***b* blocks the first channel **125***b*, and thus the bead chain will not fall out of the first channel **125***b* even when subjected to severe shaking. The snap fit between the limiting sleeve **10***b* and the fixed disk **21***b* or pivot plate **31***b* ensures that the gripper **100***b* maintains a state in which the hole wall of the fixing hole **212***b* or second adjustment hole **311***b* blocks the first channel **125***b* during use, such that the bead chain cannot fall out of the limiting sleeve **10***b* for a long time.

[0100] To ensure that the bead chain can be smoothly passed through the fixing hole **212***b* or the second adjustment hole **311***b* during the mounting process, in a conventional crane machine, the diameter of the fixing hole **212***b* or the second adjustment hole **311***b* is set to be sufficiently large, which means that the bead chain has a relatively wide range of movement. However, when the gripper tends to tilt or does tilt, the deflection of the bead chain through the fixing hole **212***b* or the second adjustment hole **311***b* may cause or exacerbate the tilting effect, resulting in a low success rate of gripping by the gripper. In addition, when the crane machine is in use, the bead chains through the fixing hole **212***b* and the first adjustment hole **211***b* often become entangled with each other. For this reason, the player will often operate the controller to cause the gripper to shake violently, such that the entangled bead chains can be unknotted. However, violent shaking may cause the bead chains to detach from the fixed disk **21***b* or the pivot plate **31***b*.

[0101] In the present embodiments, the gripper of the crane machine may solve the problem of the gripper tilting easily and the success rate of grasping objects being not high due to the oversize of the diameter of the fixing hole **212***b* or the second adjustment hole **311***b* through the setting of the limiting sleeve **10***b*. The diameter of the fixing hole **212***b* or the second adjustment hole **311***b* is greater than the maximum width of the first channel **125***b*, i.e., the maximum width of the first channel **125***b* is less than the diameter of the fixing hole **212***b* or the second adjustment hole **311***b*. When the bead chain is mounted in the limiting sleeve **10***b*, the swing range of the bead chain is limited by the width of the first channel **125***b*, which limits the swing range of the bead chain, thereby making it less likely for the gripper to become unbalanced, and thus improving the success rate of the gripper in grasping objects. Compared with the situation where the diameter of the fixing hole **212***b* or the second adjustment hole **311***b* is reduced to limit the swing range of the bead chain, the present embodiments make installation more convenient and efficient. Specifically, one end of the bead chain is connected to a driving device, and the other end is first passed through the fixing hole **212***b* or the second adjustment hole **311***b*, and then mounted in the limiting sleeve **10***b*; next, the limiting sleeve **10***b* is snapped onto the fixed disk **21***b* or the pivot plate **31***b*. In contrast, reducing the diameter of the fixing hole **212***b* or the second adjustment hole **311***b* makes it significantly more difficult to thread the bead chain through the fixing hole **212***b* or the second adjustment hole **311***b*, which significantly reduces assembly efficiency.

[0102] In addition, in the present disclosure, the relative position of one end of the bead chain to the fixed disk **21***b* or the pivot plate **31***b* is made stable by means of the end wall **12***b*, and interlocking between the limiting sleeve **10***b* and the fixed disk **21***b* or the pivot plate **31***b*, so as to prevent the bead chain from detaching from the fixed disk **21***b* or the pivot plate **31***b* when impacted by external force.

[0103] Further, an inner wall of the first channel **125***b* is arranged with an anti-disconnecting member **122***b* protruding from a middle position along its extension direction, to prevent the bead chain from disengaging from the limiting sleeve **10***b*. The anti-disconnecting member **122***b* may assist in the pre-installation of the bead chain and the limiting sleeve **10***b*, thereby preventing the bead chain from disengaging from the limiting sleeve **10***b* along the first channel **125***b* before the limiting sleeve **10***b* is mounted on the fixed disk **21***b* or the pivot plate **31***b*. The anti-disconnecting member **122***b* may include one or more pieces, and the specific number and form are not limited herein.

[0104] Further, referring to FIGS. **11**, **13**, **16** and **17**, the anti-disconnecting member **122***b* includes two limiting protrusions **1221** arranged opposite each other, and the two limiting protrusions **1221** are arranged at intervals to form a slit **123***b*; the limiting protrusions **1221** and an inner wall surface of the first channel **125***b* enclose to define a mounting hole **121***b* adapted to the bead chain; the slit **123***b* has a width that is less than the diameter of the mounting hole **121***b*, and the slit **123***b* is connected to the mounting hole **121***b* and the second channel **13***b*.

[0105] In the embodiments, the limiting protrusion **1221** and the inner wall surface of the first channel **125***b* enclose to define the mounting hole **121***b* adapted to the bead chain, such that after the bead chain is mounted, the bead at one end of the bead chain abuts against the inner surface of the end wall **12***b*, preventing the bead chain from coming off the limiting sleeve **10***b* upwards; at the same time, the chain is allowed to swing within the mounting hole **121***b*.

[0106] The slit **123***b* is connected to the mounting hole **121***b* and second channel **13***b*, providing the bead chain with a mounting path from the second channel **13***b* to the mounting hole **121***b*. The width of the slit **123***b* is less than the diameter of the mounting hole **121***b*, such that the bead chain can move in the mounting hole **121***b* under normal circumstances, but it is difficult for the bead chain to escape from the slit **123***b*, thereby effectively preventing the bead chain from accidentally disengaging from the limiting sleeve **10***b*, and further enhancing the stability and reliability of the bead chain in the limiting sleeve **10***b*. This allows the movement of the bead chain to more accurately control the closing or separation of the clamping rod **32***b*, thereby achieving precise grasping of objects and improving the success rate of grasping the objects.

[0107] In some embodiments, referring to FIGS. **13** and **17**, an inner surface of the end wall **12***b* defines a recess **124***b*, and the first channel **125***b* is disposed in the recess **124***b*.

[0108] Compared with a situation where the inner surface of the end wall **12***b* is flat, in the above embodiments, the inner surface of the end wall **12***b* is provided with the recess **124***b*, which effectively buffers the impact force generated during the movement of the bead chain and reduces the frictional loss between the bead chain and the inner surface of the end wall **12***b*. In addition, the shape and size of the recess **124***b* may play a certain positioning role on the beads, such that the position of the bead chain in the first channel **125***b* is more stable, reducing the offset of the bead chain in the length direction of the first channel **125***b*, thereby avoiding excessive offset of the crane machine, which would cause the success rate of grasping to decrease.

[0109] In some embodiments, referring to FIGS. **16** and **17**, an outer side wall of the limiting sleeve **10***b* is arranged with a first snap-fit portion **14***b*, and the fixed disk **21***b* or the pivot plate **31***b* is arranged with a second snap-fit portion that is adapted to the first snap-fit portion **14***b*; one of the first snap-fit portion **14***b* and the second snap-fit portion is a convex portion, and the other is a groove.

[0110] In the embodiments, the first snap-fit portion **14***b* and the second snap-fit portion form a stable connection between the limiting sleeve **10***b* and the fixed disk **21***b* or the pivot plate **31***b*, thereby improving the stability of the entire gripper, reducing the risk of performance degradation or failure of the gripper due to loose parts or loose connections, and thus ensuring the reliability of the crane machine during long-term use.

[0111] Furthermore, the limiting sleeve **10***b* is arranged with a second channel **13***b* and an avoidance notch **16***b* on both sides along its axial direction; the outer side wall of the limiting sleeve **10***b* includes a guide bevel **17***b*, and the avoidance notch **16***b* is disposed on the guide bevel **17***b*; the first snap-fit portion **14***b* includes at least two protrusions, and the protrusions are disposed on both sides of the second channel **13***b* or the avoidance notch **16***b*.

[0112] Interference may occur during snap-fitting of the limiting sleeve **10***b* with the fixed disk **21***b* or the pivot plate **31***b*, resulting in unsuccessful or inaccurate installation. The second channel **13***b* and the avoidance notch **16***b* cause the sidewalls of the limiting sleeve **10***b* to undergo an appropriate elastic deformation during installation, ensuring accurate installation of the limiting sleeve **10***b*. The second channel **13***b* and the avoidance notch **16***b* are arranged on both sides of the limiting sleeve **10***b* in the axial direction of the limiting sleeve **10***b*, such that the elastic deformation of the side wall of the limiting sleeve **10***b* is more uniform, and irreversible damage to the limiting sleeve **10***b* caused by the elastic deformation may be prevented. The guide bevel **17***b* serves to guide the installation. The first snap-fit portion **14***b* includes at least two protrusions, and the multi-point engagement is more stable than the single-point engagement. The protrusion may be a protruding strip extending along a peripheral direction of the limiting sleeve **10***b*, or a catch. Here, the specific shape of the protrusion is not limited.

[0113] In some embodiments, the width of the second channel **13***b* is less than the width of the first channel **125***b*. Thus, after the bead chain is mounted in the first channel **125***b*, it is restricted by the second channel **13***b* and is not easily disengaged from the limiting sleeve **10***b*, thereby assisting in the pre-installation of the limiting sleeve **10***b* with the bead chain.

[0114] The pivot structure is a key component for opening and closing the multiple clamping rods **32***b*. The pivot structure includes a fixing post **431***b* and a deformable part connected to each other, where the fixing post **431***b* and the deformable part are arranged along an axial direction of the fixing post **431***b*. The fixing post **431***b* serves as a main supporting part of the pivot structure and plays a role in fixing and connecting. The deformable part can deform when passing through the connecting rod **22***b*/fixed disk **21***b*/pivot plate **31***b*/clamping rod **32***b*, allowing it to pass through smoothly, and then restore its shape and be limited in the axial direction of the fixing post **431***b* with the connecting rod **22***b*/fixed disk **21***b*/pivot plate **31***b*/clamping rod **32***b*. The specific shape of the deformable part is not limited herein.

[0115] Compared to a pivot structure that requires fasteners such as nuts and bolts, the pivot structure of the present embodiments does not require cumbersome tightening operations using tools during assembly. When the fixing post **431***b* is simply inserted into the connecting rod **22***b*/fixed disk **21***b*/pivot plate **31***b*/clamping rod **32***b*, the deformable part will automatically lock with the connecting rod **22***b*/fixed disk **21***b*/pivot plate **31***b*/clamping rod **32***b*. In contrast, the pivot structure using nuts and bolts requires the use of tools for disassembly and assembly, which is more cumbersome, and may increase the difficulty of disassembly due to rust or damage to the nuts and bolts.

[0116] Compared with a pivot structure of the hook type, the pivot structure of the present embodiments may achieve a more precise pivoting movement. The pivot structure of the hook type may have a large gap between the hook and the pivot rod, which causes the pivoting to shake, affecting the pivoting accuracy. However, the plug-in connection of the fixing post **431***b* and the limit of an elastic catch **432***b* in the embodiments may ensure that the relative position between the fixing post **431***b* and the connecting rod **22***b*/fixed disk **21***b*/pivot plate **31***b*/clamping rod **32***b* remains stable, such that the pivoting movement can follow the designed path, thereby reducing unnecessary deviations, improving the accuracy of the pivoting movement, and in turn helping to improve the accuracy and success rate of grasping objects.

[0117] In the present disclosure, mounting may be completed without the aid of external tools due to the deformable nature of the deformable part of the pivot structure. In addition, the accuracy and success rate of the gripper of the crane machine may be improved by inserting the fixing post **431***b* to maintain the relative position between the fixing post **431***b* and the connecting rod **22***b*/fixed disk **21***b*/pivot plate **31***b*/clamping rod **32***b* stable, thereby improving the precision of the pivot.

[0118] Further, the pivot structure includes two opposing bosses **41***b*, and a pivot tab **42***b* sandwiched between the two bosses **41***b*; the bosses **41***b* and the pivot tab **42***b* each define a through hole; the deformable part includes at least two elastic catches **432***b* arranged at one end of the fixing post **431***b*, and the elastic catches **432***b* are arranged along the axial direction of the fixing post **431***b*; the fixing post **431***b* is inserted into the through hole, and the elastic catch **432***b* is engaged with the boss **41***b* in the axial direction of the fixing post **431***b*; one of the connecting rod **22***b* and the fixed disk **21***b* is arranged with two bosses **41***b* arranged opposite each other, and the other is arranged with the pivot tab **42***b*; or, one of the connecting rod **22***b* and the clamping rod **32***b* is arranged with two bosses **41***b* arranged opposite each other, and the other is arranged with the pivot tab **42***b*; or, one of the clamping rod **32***b* and the pivot plate **31***b* is arranged with two bosses **41***b* arranged opposite each other, and the other is arranged with the pivot tab **42***b*.

[0119] In the embodiments, the elastic catch **432***b* undergoes elastic deformation when passing through the through hole to allow it to pass through the through hole smoothly, and after passing through, it resumes elastic deformation and snaps with the boss **41***b*; at this time, the fixing post **431***b* is inserted in the through hole. In this way, the pivot structure can be installed without the need for external tools.

[0120] In some embodiments, the fixing post **431***b* has an internal cavity. During the injection molding process, the plastic material shrinks during cooling, a phenomenon known as shrinkage.

Shrinkage may cause dimensional deviations, surface depressions, insufficient strength, and other problems in the molded part. The design of the fixing post **431***b* with a cavity may effectively alleviate the problem of shrinkage, such that the fixing post **431***b* maintains a smooth outer surface, which in turn allows the fixing post **431***b* to rotate flexibly within the through hole, thereby improving the pivot accuracy. In addition, the cavity design may significantly reduce the weight of the fixing post **431***b*. In the pivot structure, the lighter fixing post **431***b* may reduce the inertia of movement, making the entire pivot structure more flexible and responsive during movement. This flexibility helps to improve the movement accuracy of the pivot structure, especially in devices such as the gripper of the crane machine that require rapid adjustment and frequent movement.

[0121] In some embodiments, an end of the fixing post **431***b* away from the elastic catch **432***b* extends beyond the surface of the boss **41***b* in the axial direction of the fixing post **431***b*. When the end of the fixing post **431***b* extends beyond the surface of the boss **41***b*, a more spacious gripping space is provided for the user. This design makes it easier for the user to grip the fixing post **431***b* with their fingers without being disturbed by the boss **41***b*. The overhang allows the user to exert force more easily when operating, whether rotating, pulling or pushing, and to control the force and direction more precisely.

[0122] In some embodiments, referring to FIG. **13**, the connecting rod **22***b* and the clamping rod **32***b* are rotatably connected. An end of the connecting rod **22***b* close to the clamping rod **32***b* is arranged with a first abutment protrusion **221***b*, and a second abutment protrusion **321***b* adapted to the first abutment protrusion **221***b* is arranged on the clamping rod **32***b*; the second abutment protrusion **321***b* is arranged relative to the first abutment protrusion **221***b* close to the central axis of the pivot plate **31***b*; the second abutment protrusion **321***b* is located in the movement path of the first abutment protrusion **221***b* such that the first abutment protrusion **221***b* can be rotated to abut against the second abutment protrusion **321***b*.

[0123] In the embodiments, the rotatable connection between the connecting rod **22***b* and the clamping rod **32***b* enables the clamping rod **32***b* to rotate relative to the connecting rod **22***b*, which is the basis for the opening and closing action of the gripper. The mutual fit of the first abutment protrusion **221***b* and the second abutment protrusion **321***b* allows the relative rotation angle between the connecting rod **22***b* and the clamping rod **32***b* to be precisely controlled, thereby precisely controlling the degree of opening and closing of the gripper. When the first abutment protrusion **221***b* abuts against the second abutment protrusion **321***b* during operation of the crane machine, a limit position of the opening and closing of the gripper is reached, thereby avoiding structural damage due to excessive opening and closing or the problem of being unable to effectively grasp objects, so as to improve the operating accuracy and reliability of the gripper.

[0124] In some embodiments, a first tooth is arranged on an end of the clamping rod **32***b* facing the central axis of the pivot plate **31***b*. In this way, when adjacent clamping rods **32***b* are brought together, the first teeth on the multiple clamping rods **32***b* can mesh with each other to form a tighter gripping structure, which helps to firmly grasp the object. For some objects with smooth surfaces or irregular shapes, the first teeth may provide additional friction and grip force to prevent the object from slipping during the gripping process, thereby reducing the number of gripping failures and improving the success rate of gripping objects.

[0125] In some embodiments, referring to FIGS. **10** and **11**, the crane machine includes a pallet **50***b* that is removably mounted on the tail end of the clamping rod **32***b*; the pallet **50***b* includes a third clamping surface **52***b* disposed towards the central axis of the pivot plate **31***b* and a third holding surface **51***b* disposed opposite the third clamping surface **52***b*; the third holding surface **51***b* is attached on a plane of the clamping rod **32***b* that is close to the central axis of the pivot plate **31***b*; the pallet **50***b* further includes an enclosure **53***b* that is arranged around the edge of the third clamping surface **52***b*.

[0126] In the embodiments, the pallet **50***b* is removably mounted on the tail end of the clamping rod **32***b*, providing the gripper of the crane machine with the possibility of functional expansion.

When different types of objects are required to be grabbed, different pallets **50***b* can be replaced according to actual needs, thereby improving the adaptability and flexibility of the crane machine. For example, for larger, thinner, or specially shaped objects, the gripping task may be better completed and the success rate of gripping may be improved by replacing with a suitable pallet **50***b*.

[0127] The third clamping surface **52***b* may come into direct contact with the object. By adjusting its shape and material, it may better adapt to different object surfaces and provide better gripping results. The third holding surface **51***b* is attached to the plane of the clamping rod **32***b* near the central axis of the pivot plate **31***b* and provides a foundation for the installation and support of the pallet **50***b*, ensuring that the pallet **50***b* can be stably attached to the clamping rod **32***b* during use.

[0128] The enclosure **53***b* is attached around the edge of the clamping surface, preventing objects from slipping sideways. When grasping objects that are small or tend to roll, the enclosure **53***b* may restrict the objects to the area of the pallet **50***b*, preventing them from rolling sideways during the grasping process, thereby further enhancing the grasping function of the pallet **50***b*.

[0129] In some embodiments, referring to FIG. **11**, the crane machine includes a pallet **50***b* that is removably mounted on the tail end of the clamping rod **32***b*; the pallet **50***b* includes a third clamping surface **52***b* arranged towards the central axis of the pivot plate **31***b* and a third holding surface **51***b* arranged opposite the third clamping surface **52***b*; the third holding surface **51***b* is attached on a plane of the clamping rod **32***b* that is close to the central axis of the pivot plate **31***b*; the third holding surface **51***b* is arranged with a limiting catch **511***b* and an assembly groove **512***b*; the limiting catch **511***b* and the assembly groove **512***b* are distributed along the extension direction of the clamping rod **32***b*; a rod body of the clamping rod **32***b* is engaged with the limiting catch **511***b*, and the tail end of the clamping rod **32***b* is inserted into the assembly groove **512***b*.

[0130] In the embodiments, the limiting catch **511***b* and the assembly groove **512***b* are distributed along the extension direction of the clamping rod **32***b*, such that the pallet **50***b* may be evenly stressed in the extension direction of the clamping rod **32***b*, avoiding the pallet **50***b* from loosening or falling off due to stress during the gripping process, and ensuring the safety and reliability of the gripping operation. In addition, the assembly groove **512***b* allows the tail end of the clamping rod **32***b* to be inserted, which not only further ensures the correct installation position of the pallet **50***b*, but also makes the connection between the pallet **50***b* and the clamping rod **32***b* more closely and firmly, thereby preventing the pallet **50***b* from falling off the clamping rod **32***b* when it is subjected to a large load. Furthermore, the peripheral wall of the assembly groove **512***b* may prevent an end of the pallet **50***b* from skewing due to uneven stress.

[0131] In some embodiments, referring to FIGS. **10** and **11**, the crane machine includes a pallet **50***b* that is removably mounted on the tail end of the clamping rod **32***b*, and an end of the pallet **50***b* includes a second tooth **54***b* that is protruding towards the central axis of the pivot plate **31***b*.

[0132] In the embodiments, when adjacent clamping rods **32***b* are brought together, the pallets **50***b* on the clamping rods **32***b* also move closer to each other, and the second teeth **54***b* on the pallets **50***b* can mesh with each other to form a tighter gripping structure, which helps to firmly grasp the object. For some objects with smooth surfaces or irregular shapes, the second teeth **54***b* can provide additional friction and grip, preventing the object from slipping during the gripping process, thereby reducing the number of gripping failures and improving the success rate of gripping objects.

[0133] In some embodiments, referring to FIG. **10**, a through hole **323***b* is defined on a middle portion of the clamping rod **32***b*, and the aperture of the through hole **323***b* extends along the extension direction of the clamping rod **32***b*. Thus, during transportation, nylon ties, ropes, etc., can be passed through the through hole **50***b* on each clamping rod **32***b* to cause the clamping rods **32***b* to move closer to each other and be fixed, thereby preventing the clamping rods **32***b* from colliding and being damaged due to bumps.

[0134] The present disclosure further proposes a crane machine, including a bead chain, a driving device, and a gripper. The specific structure of the gripper refers to the above embodiments. Since the crane machine adopts all the technical solutions of all the above embodiments, it has at least all the beneficial effects brought about by the technical solutions of the above embodiments, which will not be described in detail herein. The bead chain connects the gripper and the driving device; the driving device causes the fixed disk **21***b* and the pivot plate **31***b* of the gripper to move closer to or further away from each other by driving the bead chain; when the fixed disk **21***b* and the pivot plate **31***b* of the gripper move closer to each other, the clamping rods **32***b* of the gripper move closer to each other; and when the fixed disk **21***b* and the pivot plate **31***b* of the gripper move away from each other, the clamping rods **32***b* of the gripper move away from each other.

[0135] Finally, it should be noted that the above embodiments are only intended to illustrate the technical solutions of the present disclosure, and not to limit them. Although the present disclosure has been described in detail with reference to the foregoing embodiments, those skilled in the art should understand that it is still possible to make modifications to the technical solutions recorded in the foregoing embodiments, or to make equivalent replacements for some of the technical features therein. These modifications or substitutions do not cause the essence of the technical solutions to depart from the spirit and scope of the technical solutions of the embodiments in the present disclosure.

Claims

1. A claw device for a crane machine, comprising an adjustment assembly and an actuating assembly movably connected to the adjustment assembly; wherein the adjustment assembly comprises a fixed disk and at least two connecting rods spaced apart along an outer peripheral edge of the fixed disk; the actuating assembly comprises a pivot plate disposed below the fixed disk and at least two clamping rods spaced apart along an outer peripheral edge of the pivot plate; a viewing window is formed between each adjacent two of the at least two clamping rods; each clamping rod comprises a connecting segment, a fixed segment, and a holding segment that are sequentially connected; the connecting segment is movably connected to the pivot plate; the fixed segment comprises a first clamping surface toward a center axis of the pivot plate; the holding segment extends toward the center axis of the pivot plate from a lower end of the fixed segment; the holding segment comprises a first holding surface toward the center axis of the pivot plate; the first holding surface is connected to the first clamping surface, and the first holding surface and the first clamping surface are arranged at an angle; an end of each connecting rod is movably connected to the fixed disk, and the other end of the connecting rod is movably connected to the connecting segment of a corresponding clamping rod; wherein the fixed disk defines a first adjustment hole and a fixing hole, and the first adjustment hole and the fixing hole both pass through the fixed disk in an up-down direction; the pivot plate defines a second adjustment hole, and the second adjustment hole passes through the pivot plate in the up-down direction; a gravity center of the fixed disk is located in the first adjustment hole, and a projection of the gravity center of the fixed disk falls within the second adjustment hole; wherein the claw device further comprises a limiting column; the limiting column defines a limiting hole, and the limiting hole runs through the limiting column in the up-down direction; the limiting hole is arranged with a mounting notch, and a width of the mounting notch is less than an inner diameter of the limiting hole; wherein, a lower part of the fixing hole corresponds to the limiting column, and an aperture of the fixing hole is greater than an aperture of the limiting hole; or another the limiting column is arranged corresponding to a lower side of the second adjustment hole, and an aperture of the second adjustment hole is greater than an aperture of the limiting hole.

2. The claw device according to claim 1, wherein a lower surface of the fixed disk defines a first accommodation groove; a lower end of the fixing hole runs through the first accommodation groove, and the limiting column is arranged in the first accommodation groove; a sidewall surface of the first accommodation groove abuts against an outer peripheral surface of the limiting column and closes or seals the mounting notch of the limiting column.

3. The claw device according to claim 1, wherein a lower surface of the pivot plate defines a second accommodation groove; a lower end of the second adjustment hole runs through the second accommodation groove, and the other limiting column is arranged in the second accommodation groove; a sidewall surface of the second accommodation groove abuts against an outer peripheral surface of the other limiting column and closes or seals the mounting notch of the other limiting column.

4. A gripper for a crane machine, comprising: an adjustment assembly, comprising a fixed disk and at least two connecting rods arranged at intervals along an outer periphery of the fixed disk; wherein the fixed disk defines a first adjustment hole and a fixing hole, and the first adjustment hole and the fixing hole both extend along an up-down direction and through the fixed disk; the first adjustment hole and the fixing hole are configured for a bead chain to move up and down; an actuating assembly, movably connected to the adjustment assembly; wherein the actuating assembly comprises a pivot plate arranged below the fixed disk and at least two clamping rods arranged at intervals along an outer periphery of the pivot plate; a middle portion of the pivot plate defines a second adjustment hole extending along the up-down direction and through the pivot plate, and the second adjustment hole is configured for the bead chain to move up and down; an end of each connecting rod is movably connected to the fixed disk, and another end of the connecting rod is movably connected to a corresponding clamping rod; an end of each clamping rod is movably connected to the pivot plate; tail ends of each adjacent two of the at least two clamping rods are capable of moving together and of moving separated to achieve an opening-closing function of the gripper; and a limiting sleeve; wherein the limiting sleeve defines an opening on an end along an axial direction of the limiting sleeve and is arranged with an end wall at an opposite end; the end wall is arranged with a first channel, and a side wall of the limiting sleeve is arranged with a second channel communicating with the first channel; wherein the limiting sleeve is arranged below the fixing hole, a diameter of the fixing hole being greater than a maximum width of the first channel of the limiting sleeve; or, another the limiting sleeve is arranged below the second adjustment hole, a diameter of the second adjustment hole being greater than a maximum width of the first channel of the other limiting sleeve.

5. The gripper according to claim 4, wherein an inner wall of the first channel is arranged with an anti-disconnecting member protruding from a middle position along an extension direction of the first channel, to prevent the bead chain from disengaging from the limiting sleeve.

6. The gripper according to claim 5, wherein the anti-disconnecting member comprises two limiting protrusions arranged opposite each other, and the two limiting protrusions are arranged at intervals to define a slit; the two limiting protrusions and an inner wall surface of the first channel enclose to define a mounting hole adapted to the bead chain; a width of the slit is less than a diameter of the mounting hole, and the slit is connected to the mounting hole and the second channel.

7. The gripper according to claim 4, wherein an inner surface of the end wall defines a recess, and the first channel is disposed in the recess.

8. The gripper according to claim 4, wherein an outer side wall of the limiting sleeve is arranged with a first snap-fit portion, and the fixed disk or the pivot plate is arranged with a second snap-fit portion that is adapted to the first snap-fit portion; one of the first snap-fit portion and the second snap-fit portion is a convex portion, and the other of the first snap-fit portion and the second snap-fit portion is a convex portion is a groove.

9. The gripper according to claim 8, wherein the limiting sleeve is arranged with the second channel and an avoidance notch on opposite sides along the axial direction; the outer side wall of the limiting sleeve comprises a guide bevel, and the avoidance notch is disposed on the guide bevel; the first snap-fit portion comprises at least two protrusions, and the at least two protrusions

are disposed on opposite sides of the second channel or the avoidance notch.
10. The gripper according to claim 4, wherein a width of the second channel is less than a width of the first channel.
11. The gripper according to claim 4, wherein the gripper further comprises a pivot structure comprising a fixing post and a deformable part connected to each other, and the fixing post and the deformable part are arranged along an axial direction of the fixing post; wherein the fixing post and the deformable part pass through a corresponding connecting rod and the fixed disk, and the deformable part as well as the corresponding connecting rod or the fixed disk are limited in the axial direction of the fixing post; or, the fixing post and the deformable part pass through a corresponding connecting rod and a corresponding clamping rod, and the deformable part as well as the corresponding connecting rod or the corresponding clamping rod are limited in the axial direction of the fixing post; or, the fixing post and the deformable part pass through a corresponding clamping rod and the pivot plate, and the deformable part as well as the corresponding clamping rod or the pivot plate are limited in the axial direction of the fixing post.
12. The gripper according to claim 11, wherein the pivot structure further comprises two opposing bosses, and a pivot tab sandwiched between the two bosses; the two bosses and the pivot tab each define a through hole; the deformable part comprises at least two elastic catches arranged on an end of the fixing post, and the at least two elastic catches are arranged along the axial direction of the fixing post; the fixing post is inserted into the through hole, and the at least two elastic catches are engaged with the two bosses in the axial direction of the fixing post; wherein one of a corresponding connecting rod and the fixed disk is arranged with the two bosses arranged opposite each other, and the other of the corresponding connecting rod and the fixed disk is arranged with the pivot tab; or, one of a corresponding connecting rod and the clamping rod is arranged with the two bosses, and the other of the corresponding connecting rod and the clamping rod is arranged with the pivot tab; or, one of a corresponding clamping rod and the pivot plate is arranged with the two bosses, and the other of the corresponding clamping rod and the pivot plate is arranged with the pivot tab.
13. The gripper according to claim 12, wherein an end of the fixing post away from the at least two elastic catches extends beyond a surface of the two bosses in the axial direction of the fixing post.
14. The gripper according to claim 11, wherein the fixing post has an internal cavity.
15. The gripper according to claim 4, wherein each connecting rod and a corresponding clamping rod are rotatably connected; an end of the connecting rod close to the corresponding clamping rod is arranged with a first abutment protrusion, and a second abutment protrusion adapted to the first abutment protrusion is arranged on the corresponding clamping rod; the second abutment protrusion is arranged relative to the first abutment protrusion close to a central axis of the pivot plate; the second abutment protrusion is located in a movement path of the first abutment protrusion, and the first abutment protrusion is caused to be rotatable to abut against the second abutment protrusion.
16. The gripper according to claim 4, wherein the gripper further comprises a pallet that is removably mounted on the tail end of each clamping rod; the pallet comprises a third clamping surface disposed towards a central axis of the pivot plate and a third holding surface disposed opposite the third clamping surface; the third holding surface is attached on a plane of the clamping rod that is close to the central axis of the pivot plate; the pallet further comprises an enclosure that is arranged around an edge of the third clamping surface.
17. The gripper according to claim 4, wherein the gripper further comprises a pallet that is removably mounted on the tail end of each clamping rod; the pallet comprises a third clamping surface disposed towards a central axis of the pivot plate and a third holding surface disposed opposite the third clamping surface; the third holding surface is attached on a plane of the clamping rod that is close to the central axis of the pivot plate; the third holding surface is arranged with a limiting catch and an assembly groove; the limiting catch and the assembly groove are distributed

along an extension direction of the clamping rod; a rod body of the clamping rod is engaged with the limiting catch, and the tail end of the clamping rod is inserted into the assembly groove.

18. The gripper according to claim 4, wherein the gripper further comprises a pallet that is removably mounted on the tail end of each clamping rod; an end of the pallet comprises a second tooth that is protruding towards a central axis of the pivot plate.

19. The gripper according to claim 4, wherein a through hole is defined on a middle portion of each clamping rod, and an aperture of the through hole extends along an extension direction of the clamping rod.

20. A crane machine, comprising a bead chain, a driving device, and the gripper according to claim 4; wherein the bead chain connects the gripper and the driving device; the driving device is configured to cause the fixed disk and the pivot plate of the gripper to move closer to or away from each other by driving the bead chain; in a case where the fixed disk and the pivot plate of the gripper move closer to each other, the at least two clamping rods of the gripper move closer to each other; in a case where the fixed disk and the pivot plate of the gripper move away from each other, the at least two clamping rods of the gripper move away from each other.